

Arnav Aggarwal

B.Tech Computer Science | AI & Full-Stack Development

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 Chandigarh, India

SUMMARY

B.Tech. Computer Science student at Amity University Punjab, specialising in AI and full-stack development. Delivered an end-to-end AI-powered interview platform during a professional internship and achieved **98.4% accuracy** on a published ML research project. Published researcher with work accepted at international conferences. Passionate about solving meaningful problems at the intersection of AI, software engineering, and human experience.

PROJECTS

ARIA — AI-Assisted Interview Assessment Completed

React.js • Google Gemini API • LangChain • face-api.js • Redis • Firebase

- Built full-stack React.js platform for AI-autonomous, resume-aware interviews aligned to job description & role seniority
- Integrated Gemini API + LangChain for dynamic question generation; face-api.js for real-time gaze monitoring
- Managed stateful sessions with Redis; secured data via Firebase with automated scoring & interview logging

 **Key Learnings:** State management in stateless APIs using Redis, secure API design with Firebase Auth, real-time proctoring with computer vision, end-to-end product lifecycle.

Real-Time Indian Sign Language Recognition 98.4% Accuracy

Python • MediaPipe • OpenCV • XGBoost • ML/DL Models

- Built real-time ISL recognition pipeline using MediaPipe, OpenCV, and XGBoost for webcam gesture classification
- Designed landmark-based feature extraction using 3D hand coordinates with normalization & feature engineering
- Evaluated SVM, Random Forest, CNN, MLP models — achieved **98.4% accuracy** with explainable AI analysis

 **Key Learnings:** Computer vision pipelines, feature engineering for gesture recognition, model optimization, explainable AI techniques.

TECHNICAL SKILLS

LANGUAGES

Python C++ Java

FRAMEWORKS & LIBRARIES

React.js LangChain MediaPipe OpenCV XGBoost

TOOLS & PLATFORMS

Git Firebase Redis Google Gemini API

SOFT SKILLS

Problem-Solving Communication Time Management Teamwork

EDUCATION

B.Tech. Computer Science & Engineering

Amity University Punjab, Mohali

Aug 2023 – Present

CGPA: 7.59/10

Class XII (CBSE)

G N Khalsa Secondary School, Chandigarh

2023

69.2%

Class X (CBSE)

The Gurukul, Panchkula

2021

90.2%

EXPERIENCE

Software and AI Intern Jun – Jul 2025

Amity University Punjab, Mohali

- Independently delivered **ARIA** — full-stack AI interview platform — as primary intern deliverable in 2 months
- Worked across full product lifecycle: system design, AI integration, real-time proctoring, session management, automated scoring

KEY HIGHLIGHTS

-  **98.4% Accuracy** on ISL Recognition ML research project
-  **Published Researcher** — papers at MPCON 2026 & IEEE Conference
-  **Full Product Delivery** — End-to-end AI platform in 2 months
-  **Full-Stack + AI** — React, Python, LangChain, CV expertise

PUBLICATIONS

"Human-like Hiring at Scale: A Conversational AI and Computer Vision Approach"

Presented at MPCON 2026

"Explainable and Efficient ML Framework for Real-Time ISL Recognition Using Hand Landmark Features"

Submitted to IEEE Chandigarh Subsection Intl. Conference 2026

LANGUAGES

English (Fluent) • Hindi (Native)

INTERESTS

COURSEWORK

Data Structures & Algorithms • Object-Oriented Programming • Database Management • Artificial Intelligence • Machine Learning • Computer Vision • Software Engineering

Artificial Intelligence

Full-Stack Development

Computer Vision

Accessibility Tech

Research